CS1632:
CI / CD Pipelines

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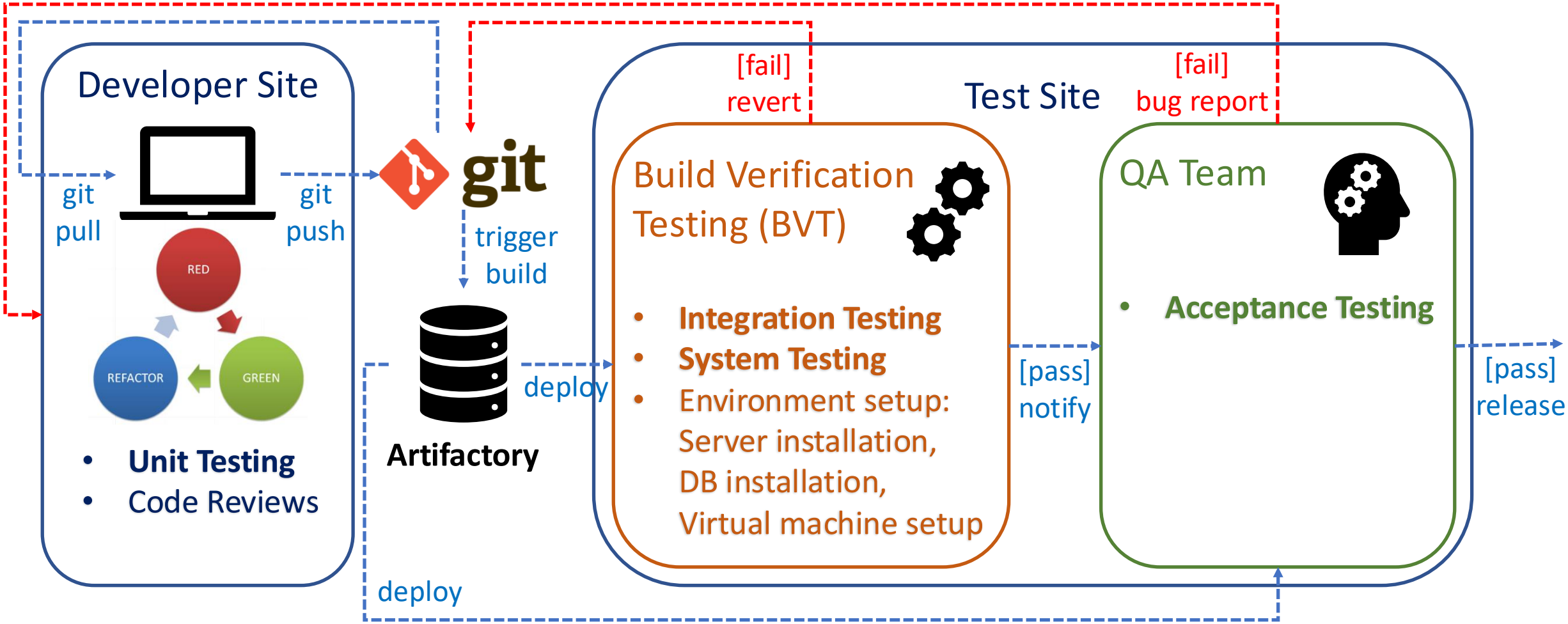
A Typical Software Testing Pipeline



When is BVT Run?

- BVT is a form of regression test
  - Checks whether program has regressed due to recent change(s)

1. When new code has been committed and a new build created
   - Verifies every single commit – ethos of continuous integration (CI)

2. Periodically (e.g. every night)
   - Verifies program hasn’t regressed during the day

What goes on in a BVT?

- Integration / system tests
  - All modules are integrated for testing (no more mocks!)
  - Typically done on an integration server, inside a virtual machine
  - In the virtual machine:
    - Deploy build from build artifactory
    - Install other software or database required by program
- (Re-)run unit tests that developer should have already run for TDD
  - To catch developers who were too lazy to run them before committing!
- Basic code quality control
  - Linters, bug finders, compilation with full warnings, …

What happens on BVT Pass / Fail?

- On BVT Pass
  - Notify QA team by (automated) email for further testing

- On BVT Fail
  - That means code repository is in a broken state now!
  - Nobody can add code to the repository until something is done about it.
  - It’s the developer’s responsibility to do one of two things:
    1. Immediately revert repository to known safe version
       (May need to revert all downstream changes along with your change, annoying many)
    2. Immediately patch the bug and run BVT again

BVT Needs to be Fast!

- BVT is in the critical path of development
  - On BVT fail, may need to revert all changes done while BVT was running
  - On BVT fail, no commit can happen until BVT passes again
  - Every minute you save in BVT is a minute you save in the development cycle

- Recommended BVT time is ~ 10 minutes
  - A lot of care needs to be put into the test cases you select for BVT
  - You should use the time to cover all major features of your program